



The Role of Information under the Risk of Ecological Collapse

EDAI Workshop @ ECAI 2025

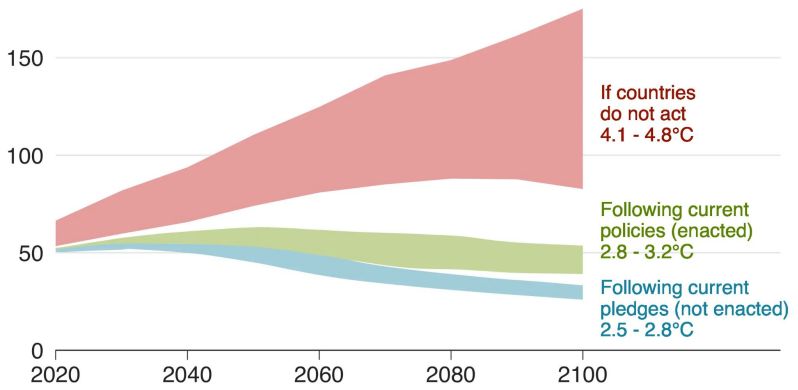
Akilan Ramakrishnan, Wolfram Barfuss



Climate change as a Collective Action Problem

How much worse will the problem get?

Emissions* and expected warming by 2100



*Emissions are in Gigatonnes of CO2 equivalent

Source: Climate Action Tracker



Source: BBC



Climate change as a Collective Action Problem



Cooperate

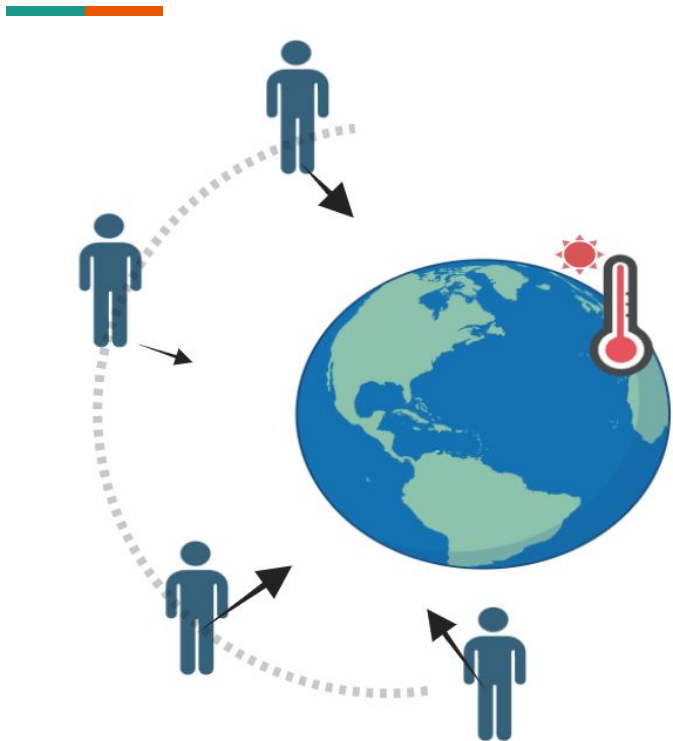
- Mitigation cost
- Keeps us within planetary boundaries



Defect

- Greater short-term profits
- Risk of Ecological Collapse with Large-Scale Impacts

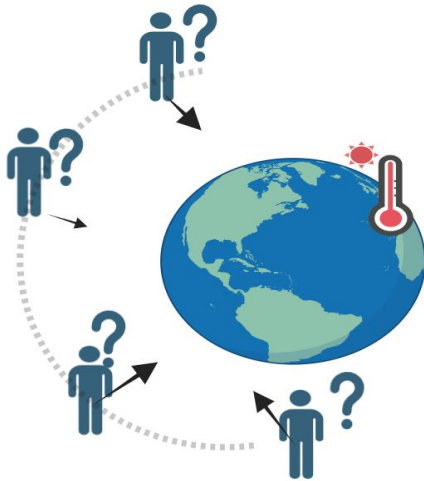
Climate change as a Collective Action Problem



- Game Theoretic Models and Laboratory Experiments (*Tavoni and Iris 2020*)

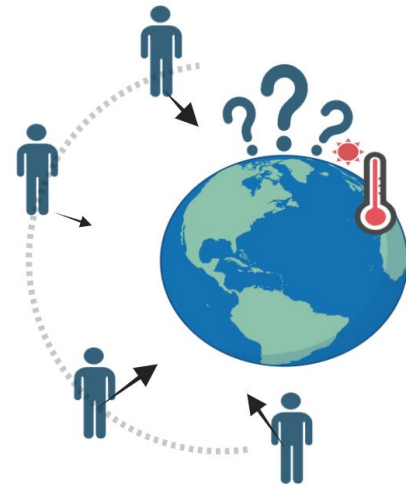
Background

Incomplete Social Information



Uncertainty about individual responses

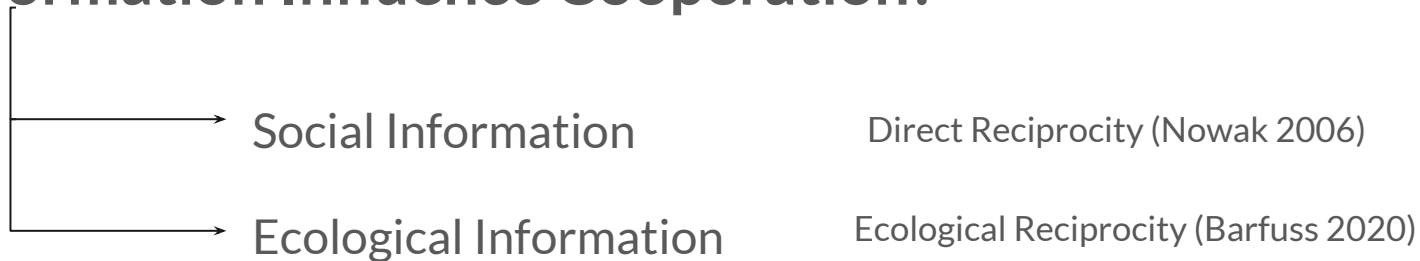
Incomplete Ecological Information



Lack of environmental awareness



How does Information Influence Cooperation?



Hypothesis: Social + Ecological Promotes Max. Cooperation

Ecological Public Goods Game



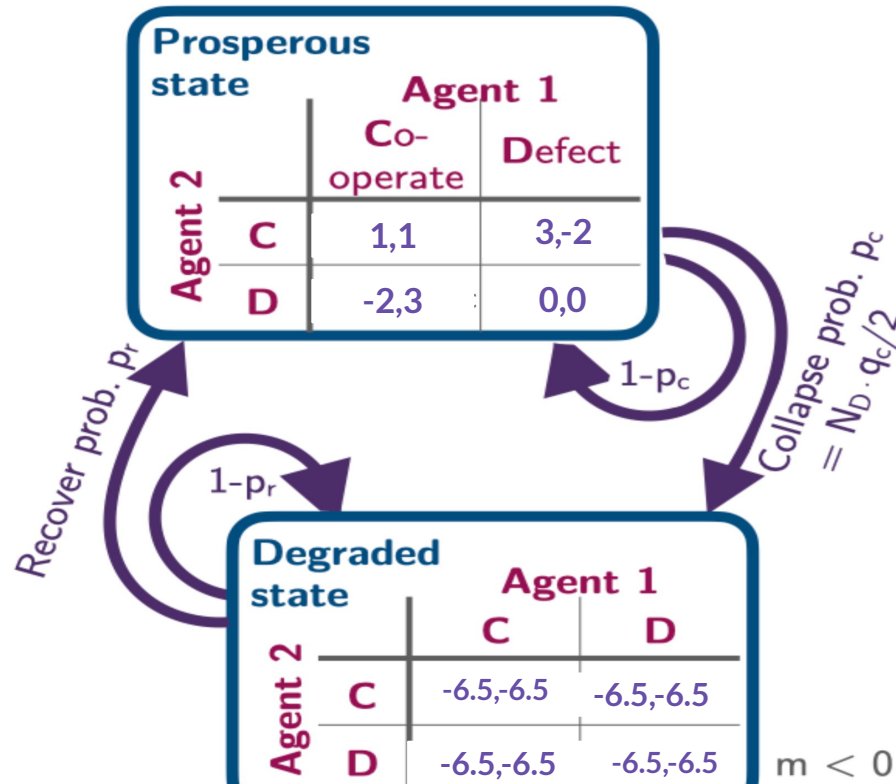
Prosperous state		Agent 1	
		Co-operate	Defect
Agent 2	C	$f \cdot C - c, f \cdot C - c$	$f \cdot C/2 - c, f \cdot C/2$
	D	$f \cdot C/2, f \cdot C/2 - c$	0,0

- Enhancement factor (f) = 1.2
- Cost (c) = 5
- 2 agents

- $p_c = 0.2$
- $p_r = 0.0001$



Ecological Public Goods Game

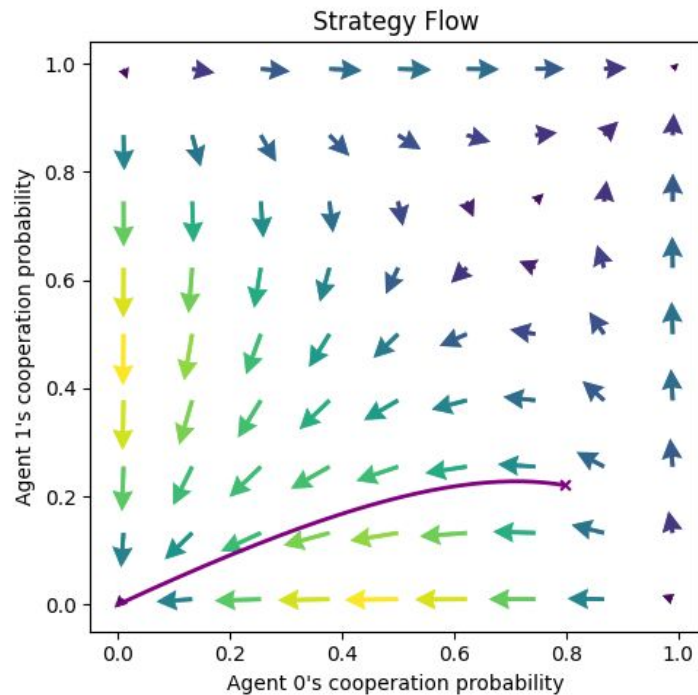




Learning Dynamics

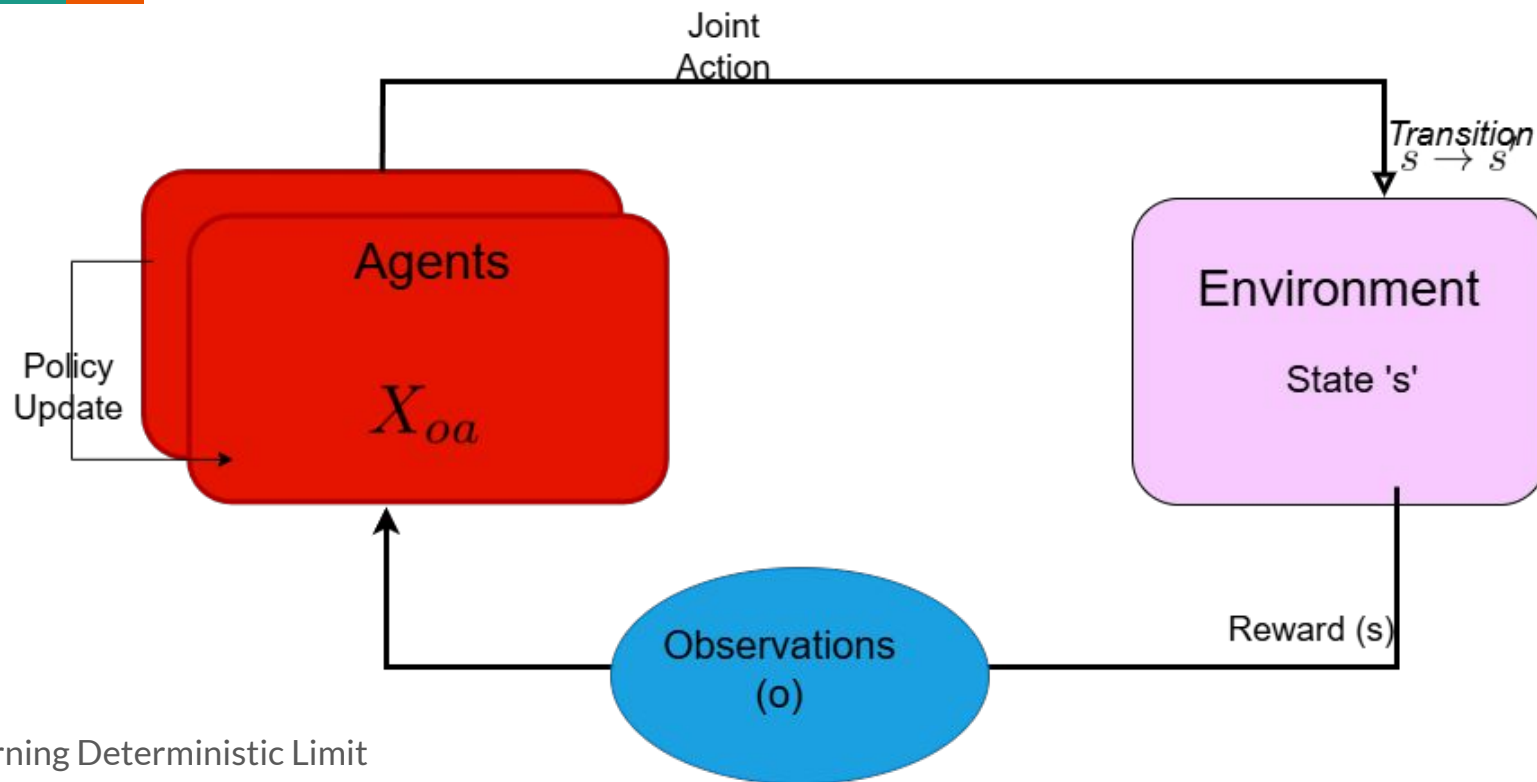
- Agents adapt their strategies based on feedback
- Multi-Agent Reinforcement Learning

Learning Dynamics



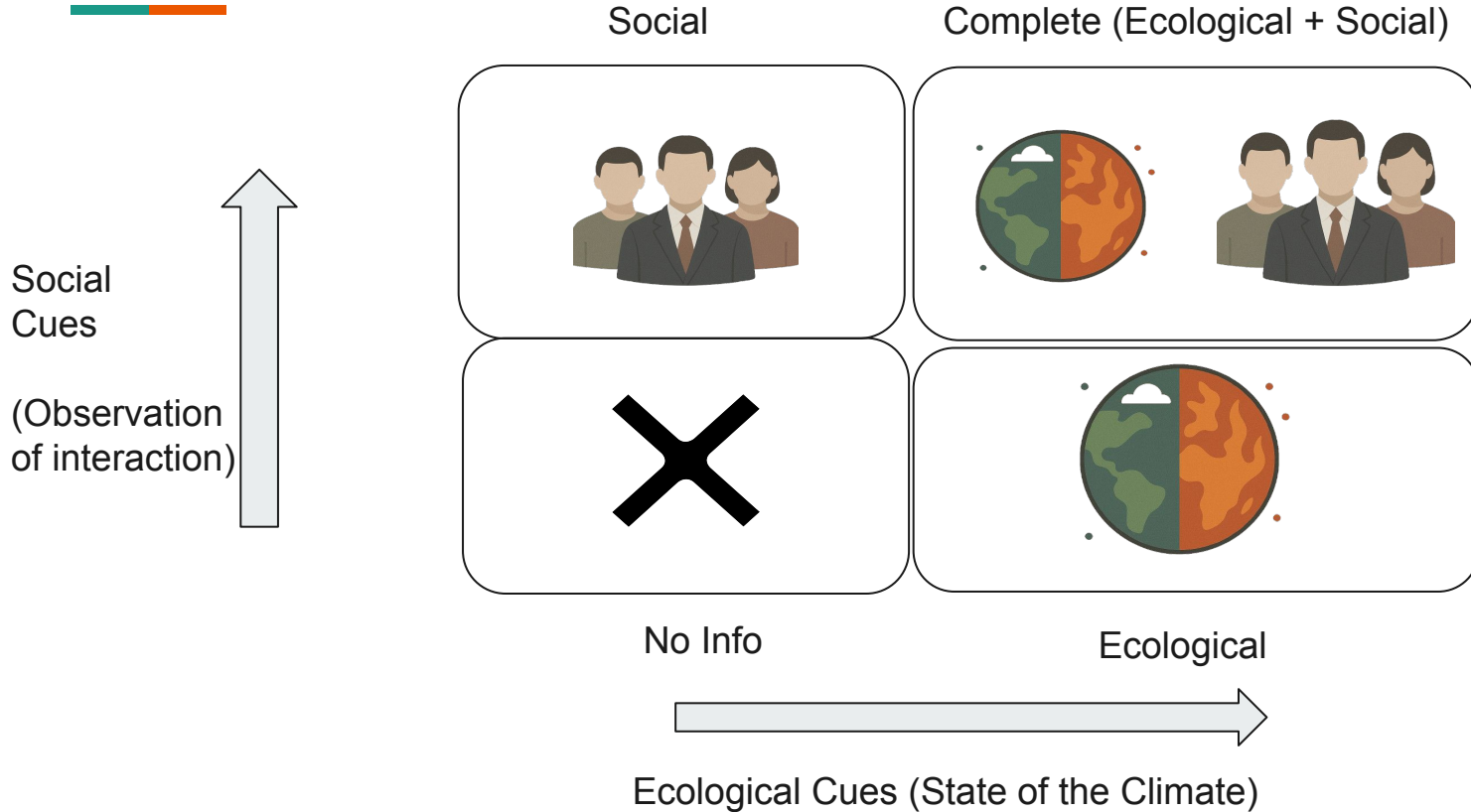
Learning Trajectory Example

MARL with Partial Observability



TD Learning Deterministic Limit

Treatments



Information Regimes

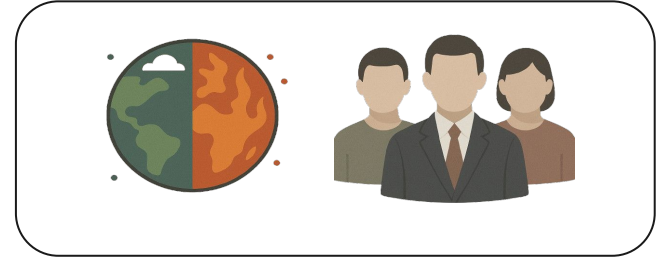


- **Ecological Info** → Current Condition (prosperous/degraded)
- **Social info** → Observation of Previous Interaction
 - Agent 1's Previous Action
 - Agent 2's Previous Action

(Memory-1)

Complete Information

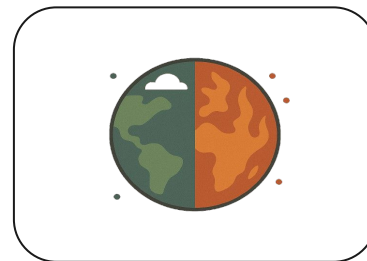
	Cooperate	Defect
p, C, D	$P(C \mid \cdot)$	$P(D \mid \cdot)$
p, D, C	$P(C \mid \cdot)$	$P(D \mid \cdot)$
p, C, C	$P(C \mid \cdot)$	$P(D \mid \cdot)$
p, D, D	$P(C \mid \cdot)$	$P(D \mid \cdot)$
g, C, D	$P(C \mid \cdot)$	$P(D \mid \cdot)$
g, D, C	$P(C \mid \cdot)$	$P(D \mid \cdot)$
g, C, C	$P(C \mid \cdot)$	$P(D \mid \cdot)$
g, D, D	$P(C \mid \cdot)$	$P(D \mid \cdot)$



- $p \rightarrow$ prosperous
- $g \rightarrow$ degraded
- 4 strategies (mem1) 2 states

Agent's Strategy under Complete Information

Ecological Information

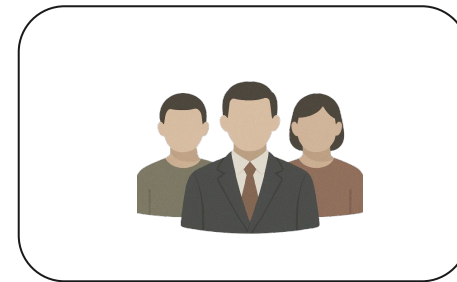


	Cooperate	Defect
p	$P(C .)$	$P(D .)$
g	$P(C .)$	$P(D .)$

- Strategy across each ecological condition

Agent's Strategy under only Ecological Information

Social Information

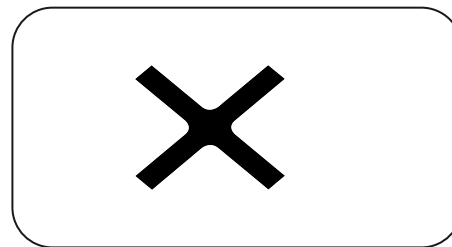


	Cooperate	Defect
C, D	$P(C \mid \cdot)$	$P(D \mid \cdot)$
D, C	$P(C \mid \cdot)$	$P(D \mid \cdot)$
C, C	$P(C \mid \cdot)$	$P(D \mid \cdot)$
D, D	$P(C \mid \cdot)$	$P(D \mid \cdot)$

- 4 strategies (Memory-1)

Agent's Strategy under only Social Information

No Information



	Cooperate	Defect
O_0	$P(C \mid .)$	$P(D \mid .)$

Agent's Strategy under No Information

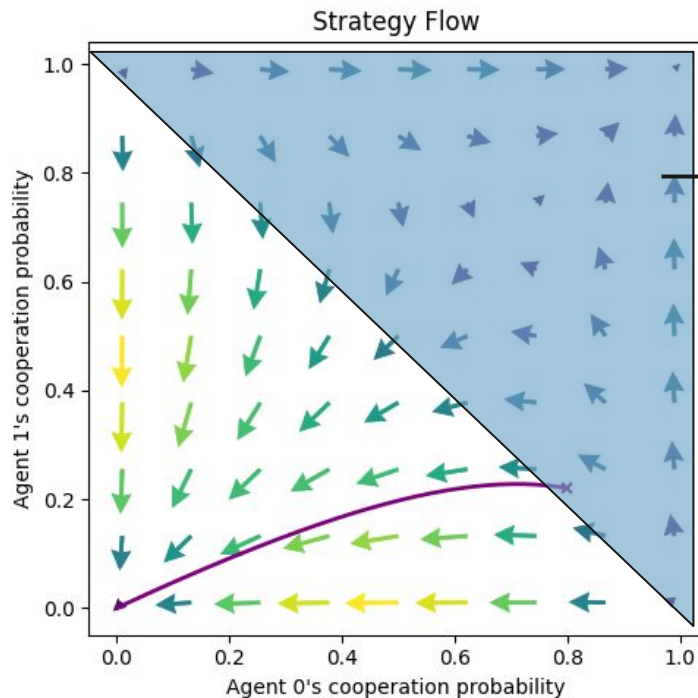
- Single indistinguishable state



Simulation Results

- Cooperation Levels: Basin of Attraction

Evaluating Cooperation Levels



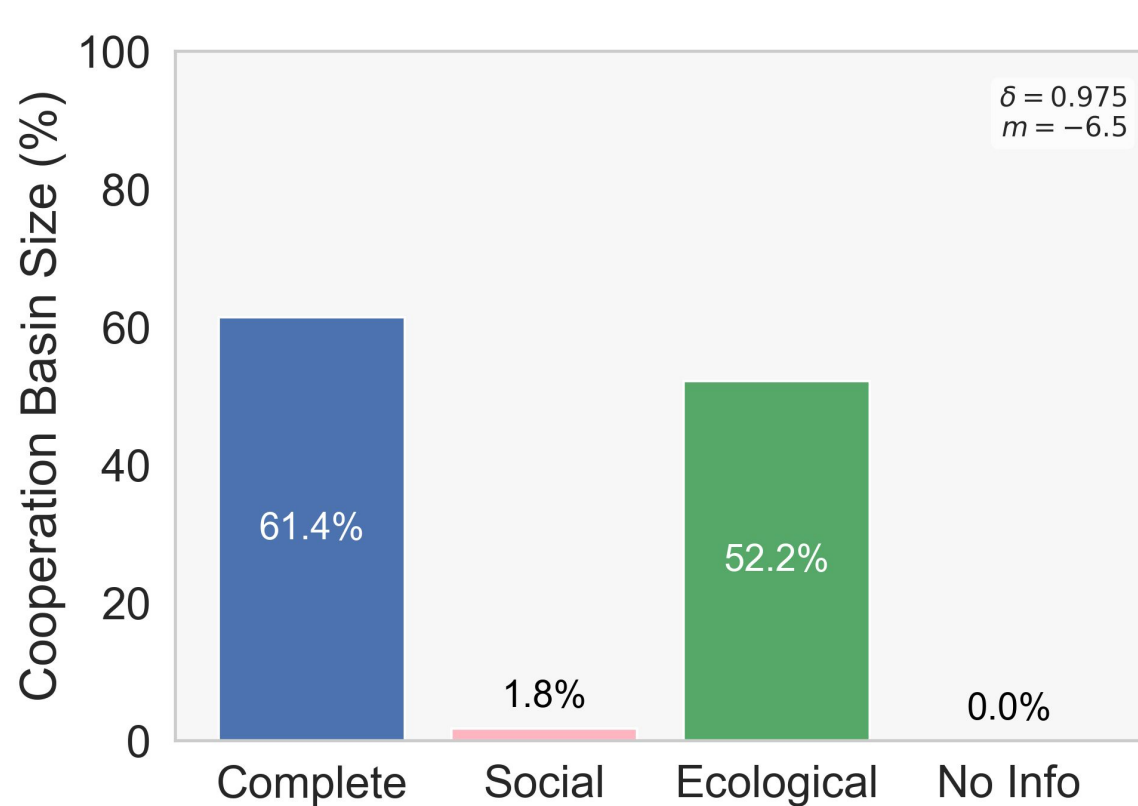
Basin of Attraction for Mutual Cooperation

- Weighted Average Cooperation across Multi-Dimensional Space

* Only Prosperous States

Basin of Attraction for Cooperation

Comparison of Info Treatments



Role of Foresight

Caring for the future can turn tragedy into comedy for long-term collective action under risk of collapse

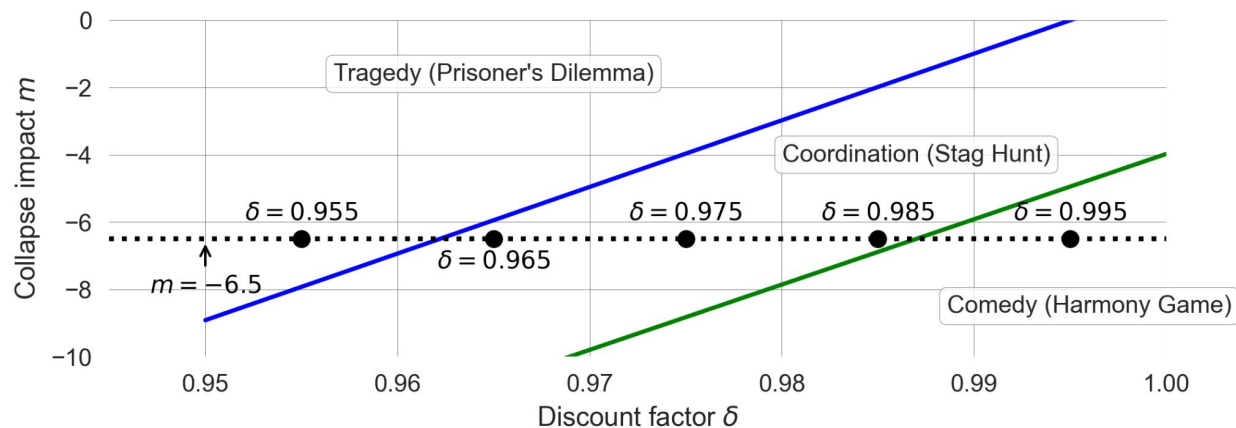
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Contributed by Simon A. Levin, March 19, 2020 (sent for review October 11, 2019; reviewed by Therese Lindahl and Arne Traulsen)

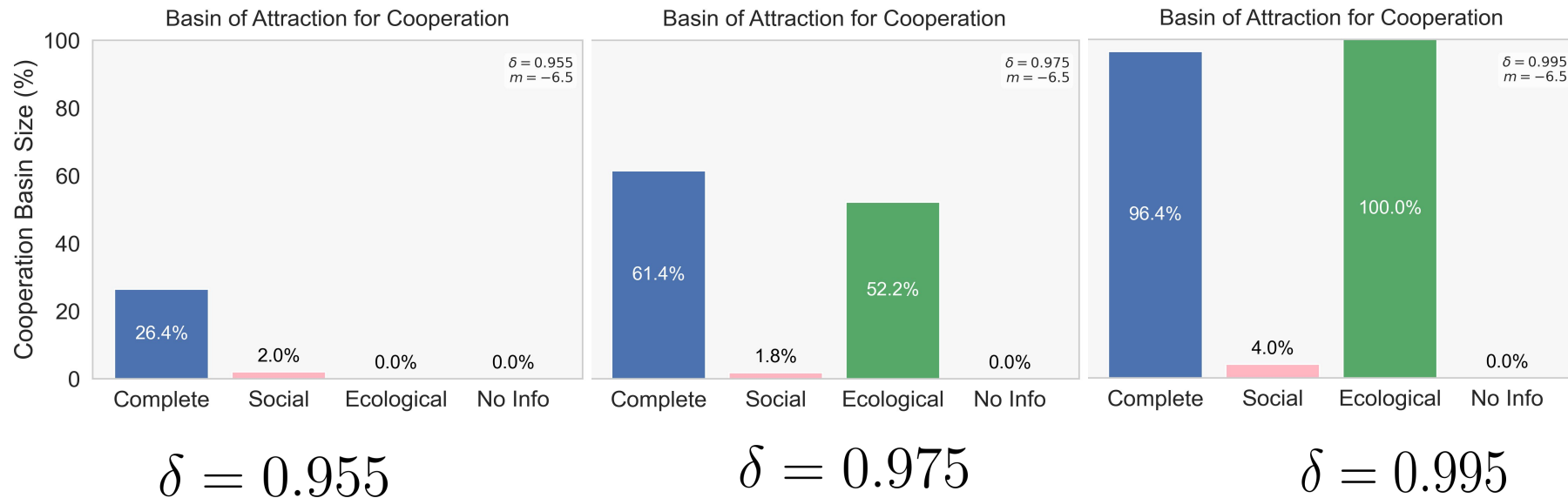
May 20, 2020 | 117 (23) 12915-12922 | <https://doi.org/10.1073/pnas.1916545117>

Barfuss 2020

$\delta \rightarrow$ discount factor :
Long-term vs Immediate



Effect on Foresight on Cooperation

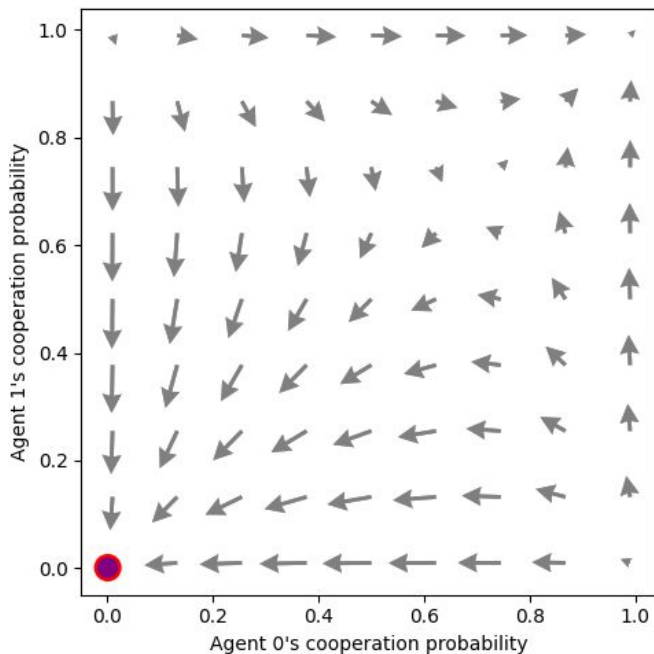




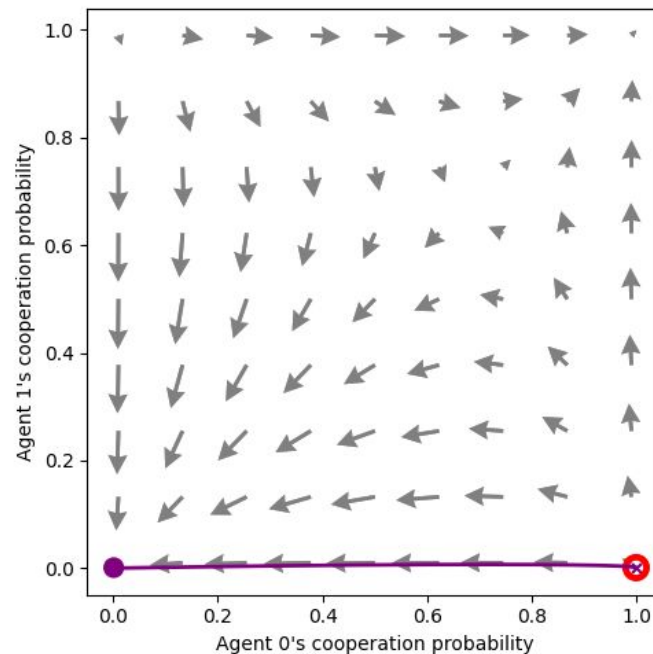
Simulation Results

- Strategies
 - Local Stability Analysis
 - Strategy Abundance

Local Stability Analysis

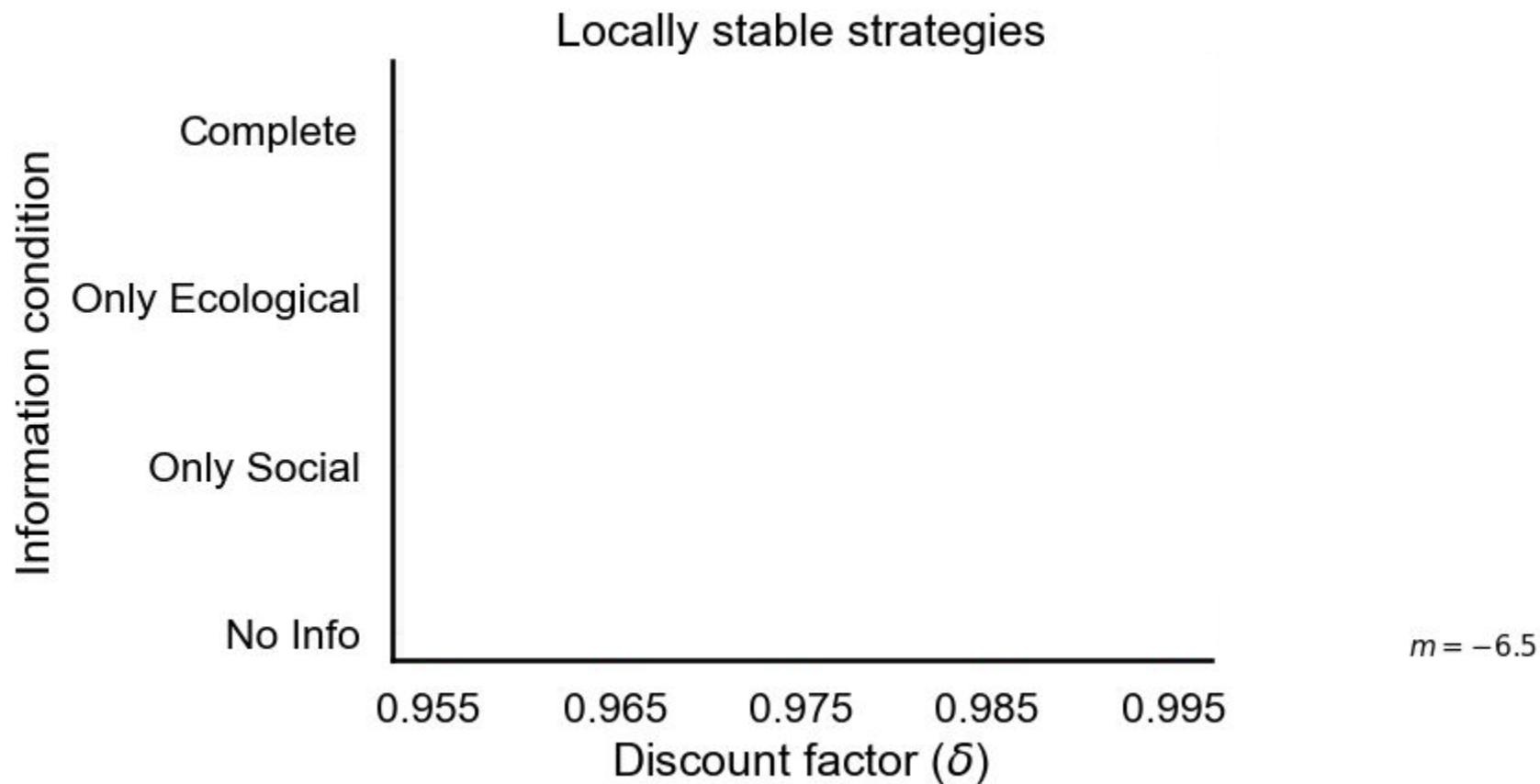


Stable Trajectory



Unstable Trajectory

Results - Local Stability Analysis



Results - Local Stability Analysis

● Cooperative ● Defective

- WSLS: Win-Stay Lose Shift
- Alternate b/w Mutual Cooperation and Defection

Strategy



ALL D



ALL C

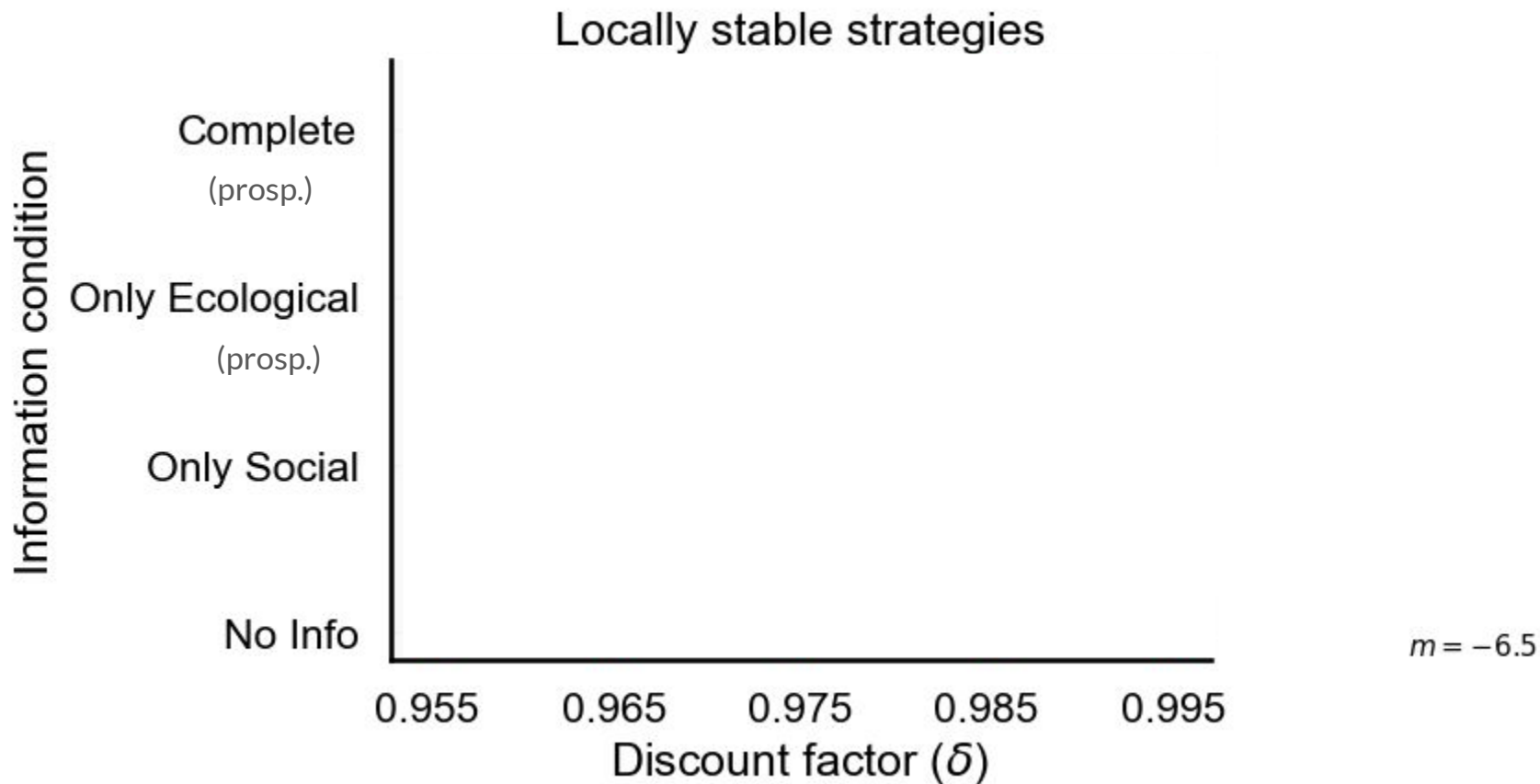


WSLS

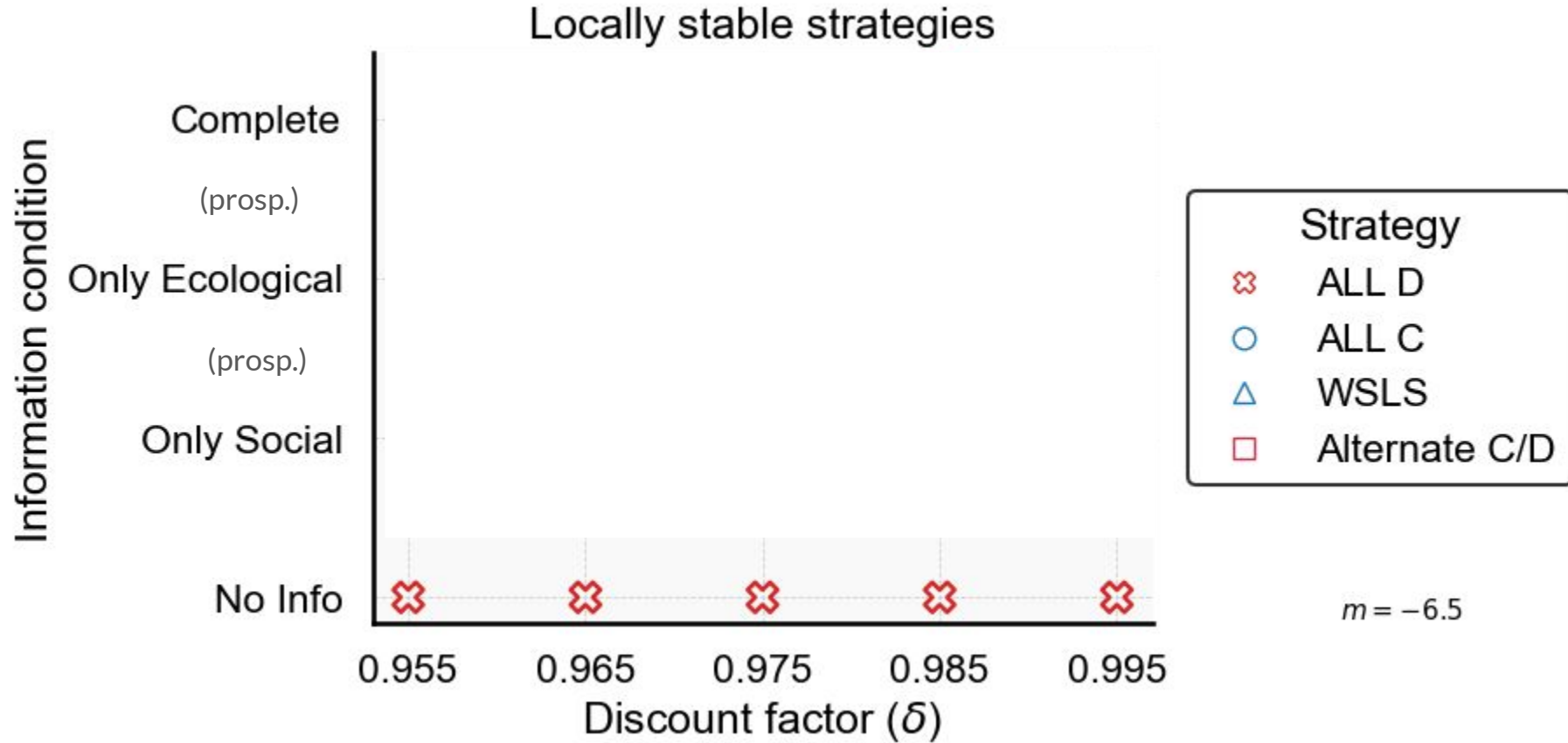


Alternate C/D

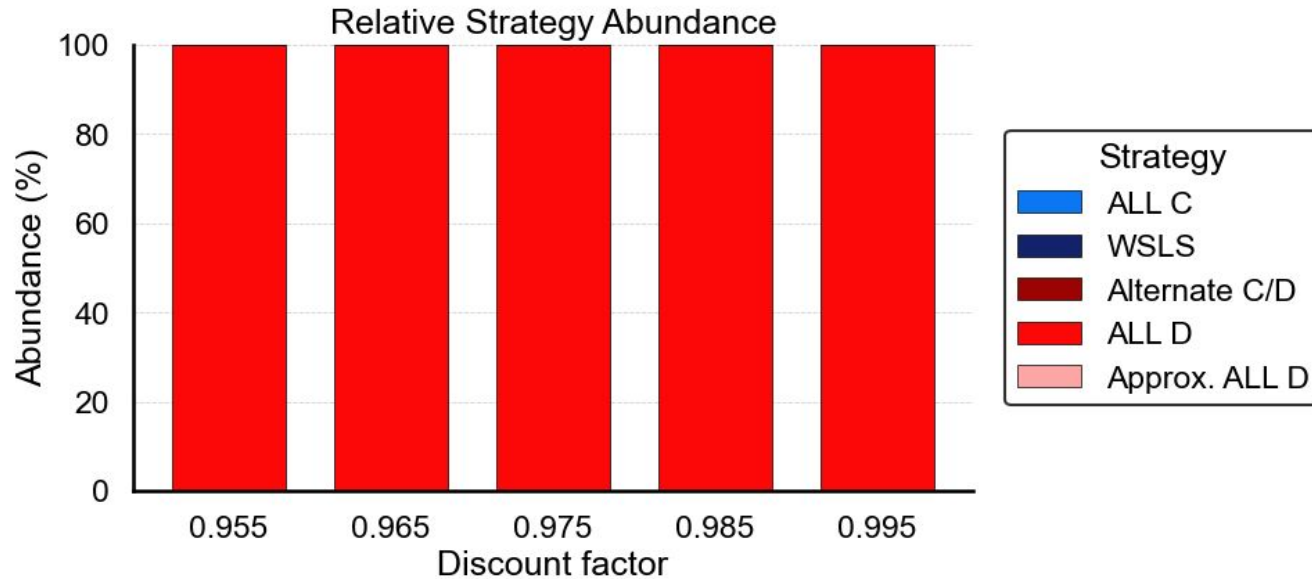
Results - Local Stability Analysis



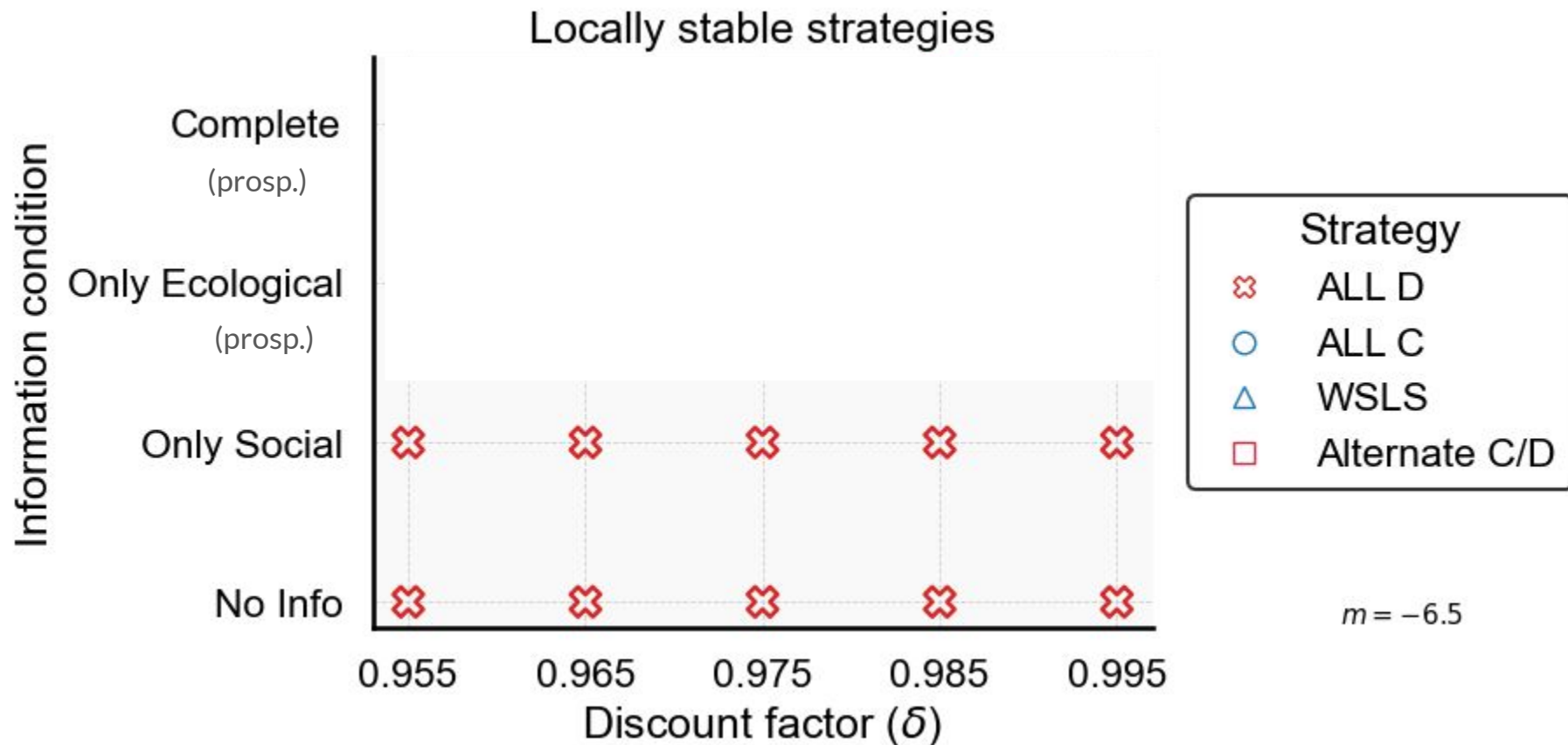
Results - Local Stability Analysis



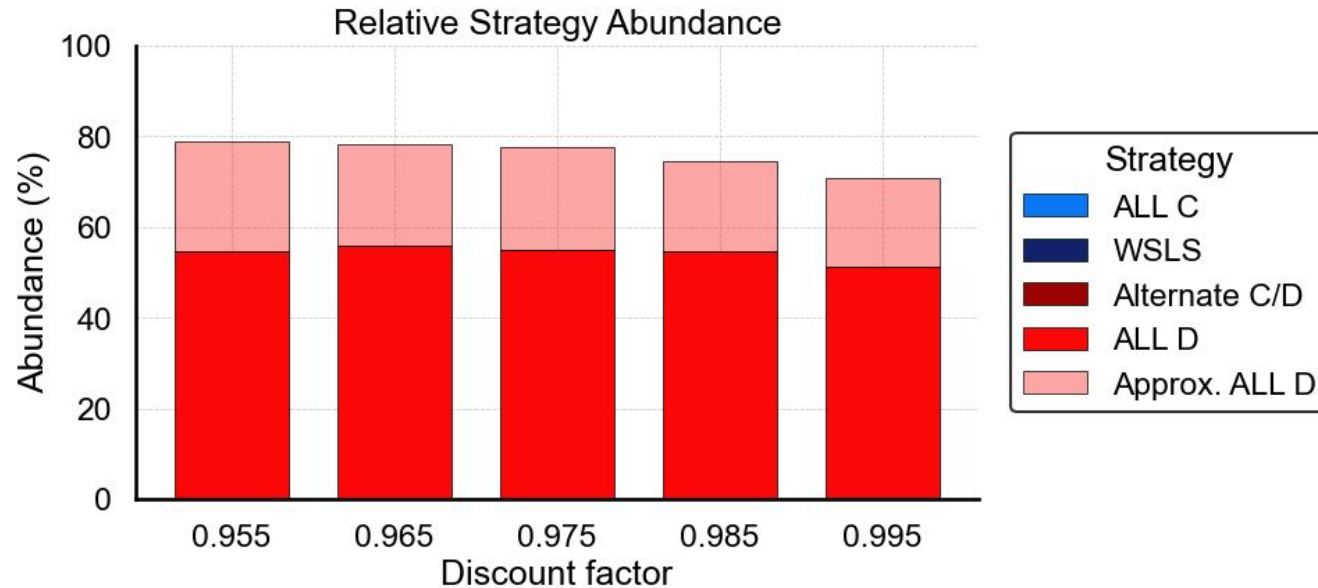
Strategy Abundance - 'No Info' Treatment



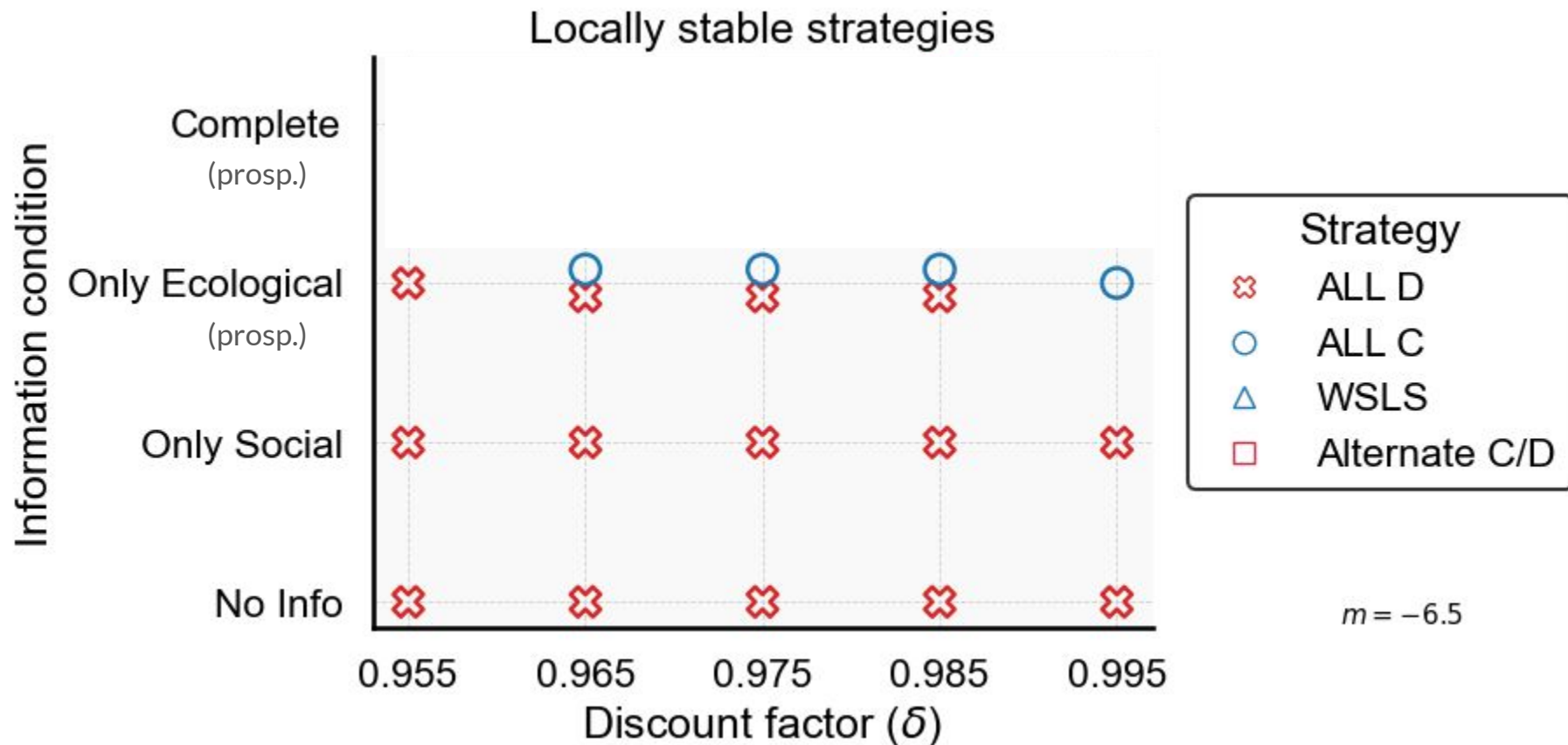
Results - Local Stability Analysis



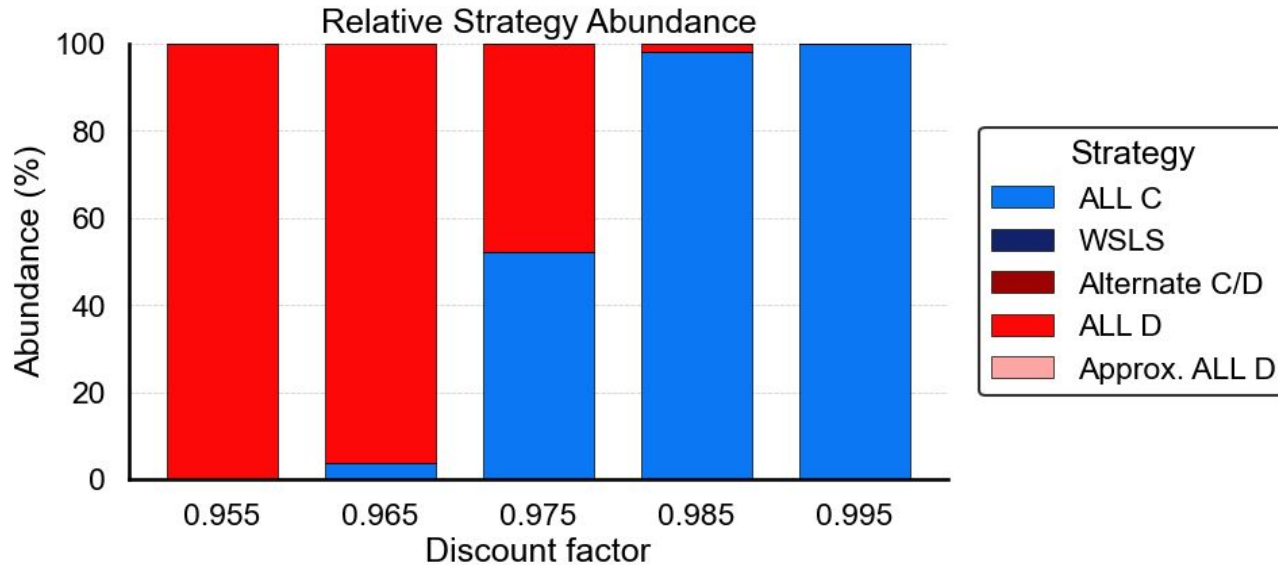
Strategy Abundance - 'Only Social' Treatment



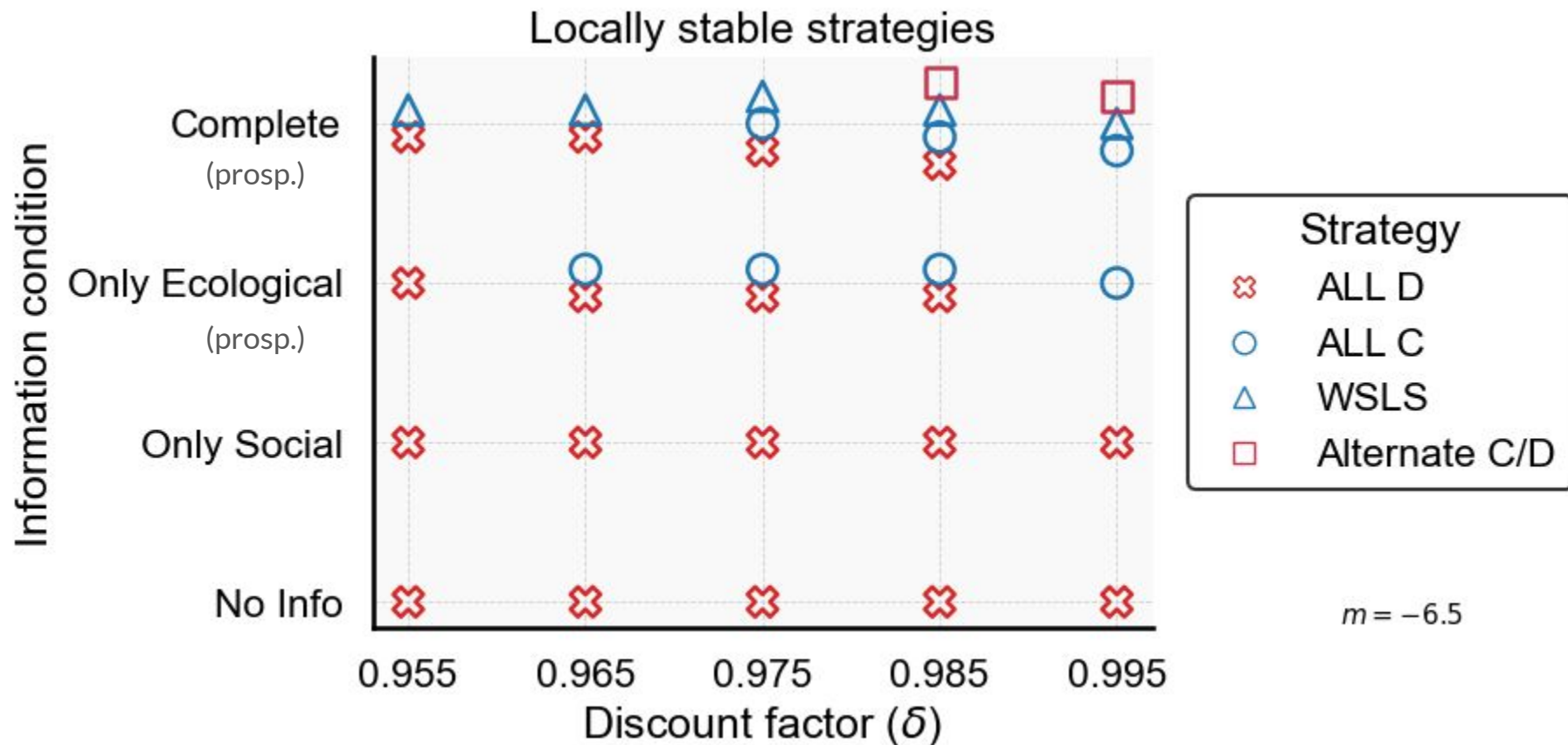
Results - Local Stability Analysis



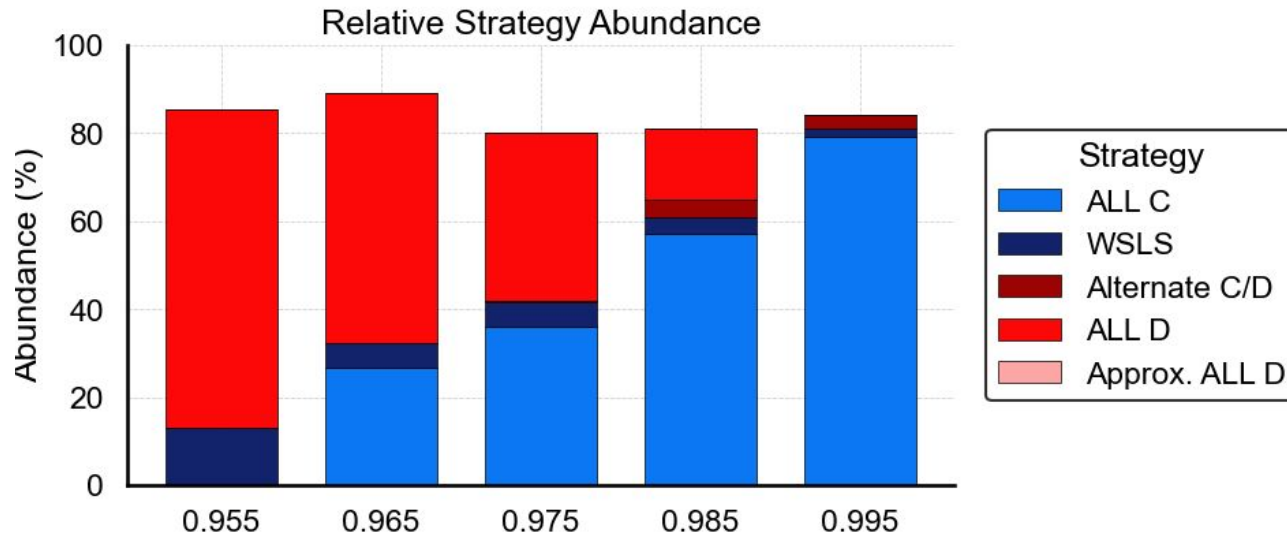
Strategy Abundance - 'Only Ecological' Treatment



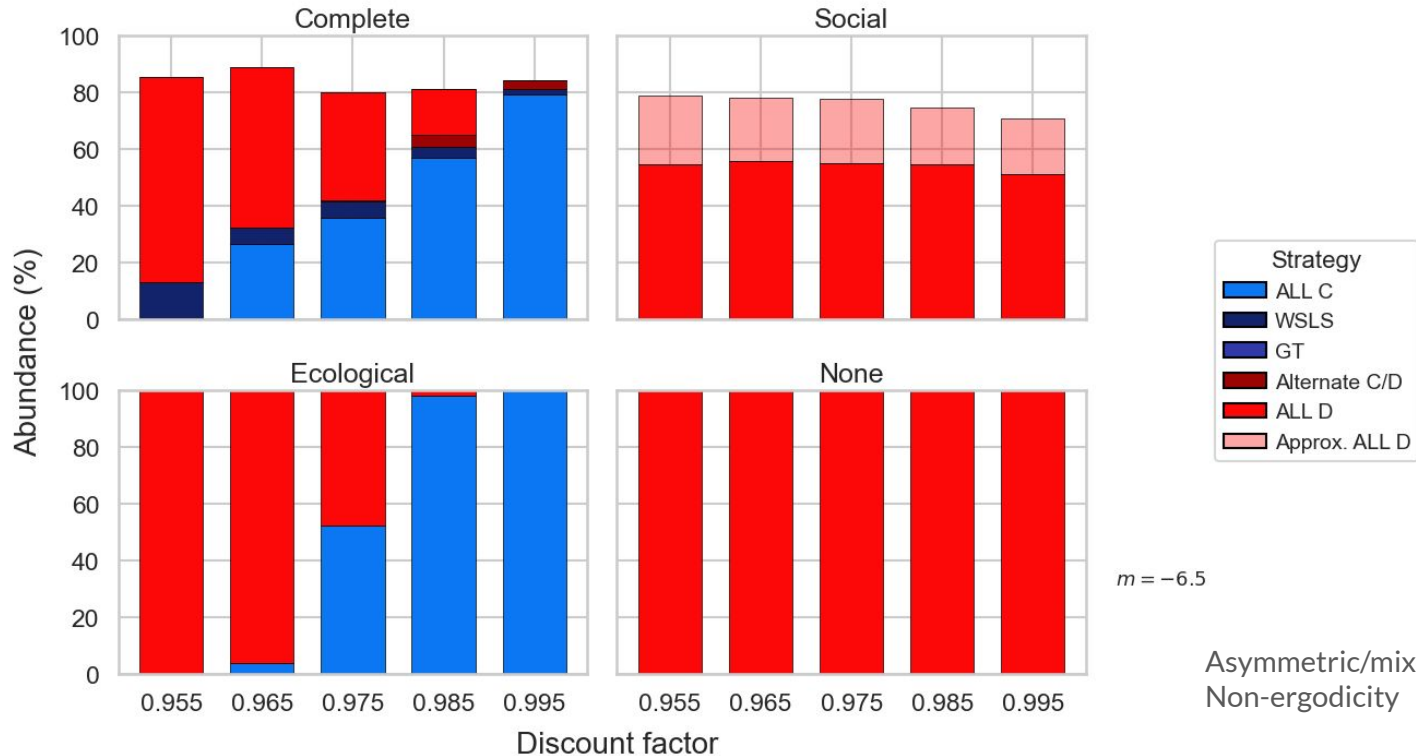
Results - Local Stability Analysis



Strategy Abundance - 'Complete' Treatment



Strategy Abundance



Summary of Findings

- 
- No Information, no cooperation
 - Social cues largely ineffective alone (for given incentive structures)
 - Ecological Information:
 - If myopic, no.
 - If sufficiently farsighted, yes
 - Social information in addition to ecological
 - Myopic: Addition of Social Info beneficial
 - Farsighted: Addition of social info can be slightly detrimental

Future work: 1) Examining different public goods parameters 2) Studying results for $N > 2$.



Thank you!

Acknowledgements



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Integrated Systems Modelling Group @ ZEF, Uni-Bonn



Conclusion

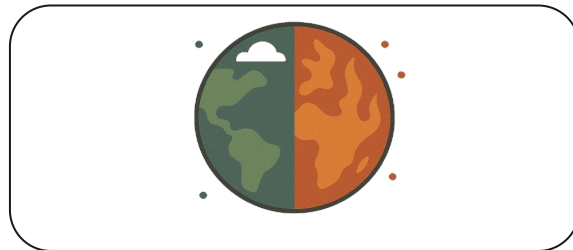
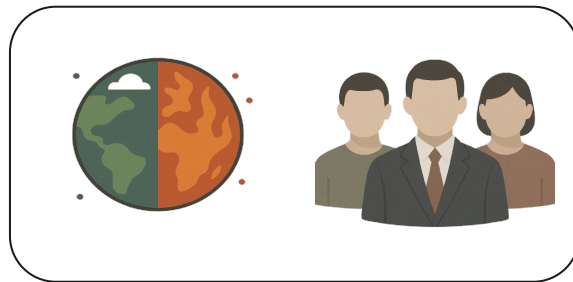
Social + Ecological Info

- Context Dependence
- Myopic: Addition of Social Info beneficial
 - Farsighted: Addition of social info harmful

Social
Cues



Both Ecological & Social Info Regime

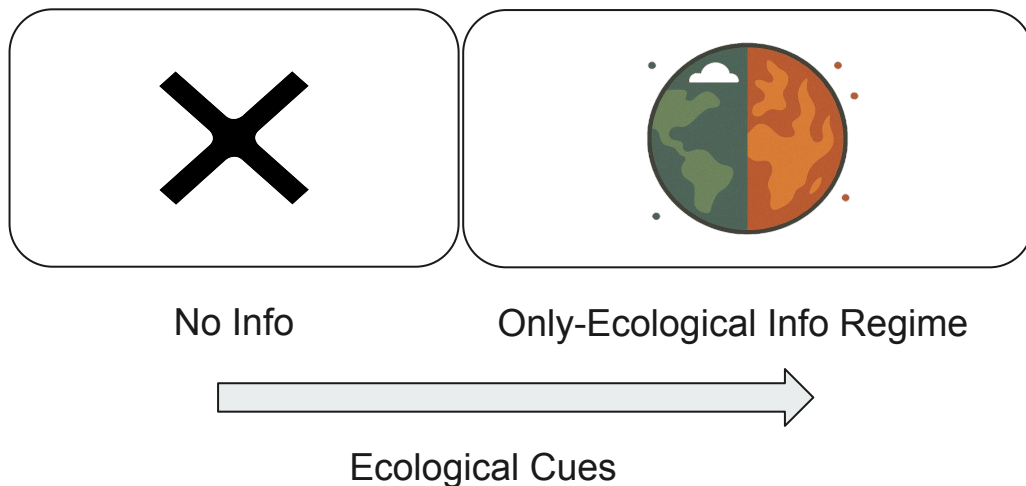


Only-Ecological Info Regime

Bistability when both cues are available

Ecological Info

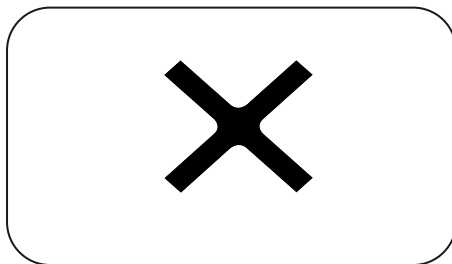
- When myopic : Does not promote cooperation
- If farsighted : Can promote cooperation; tragedy → comedy



No Info



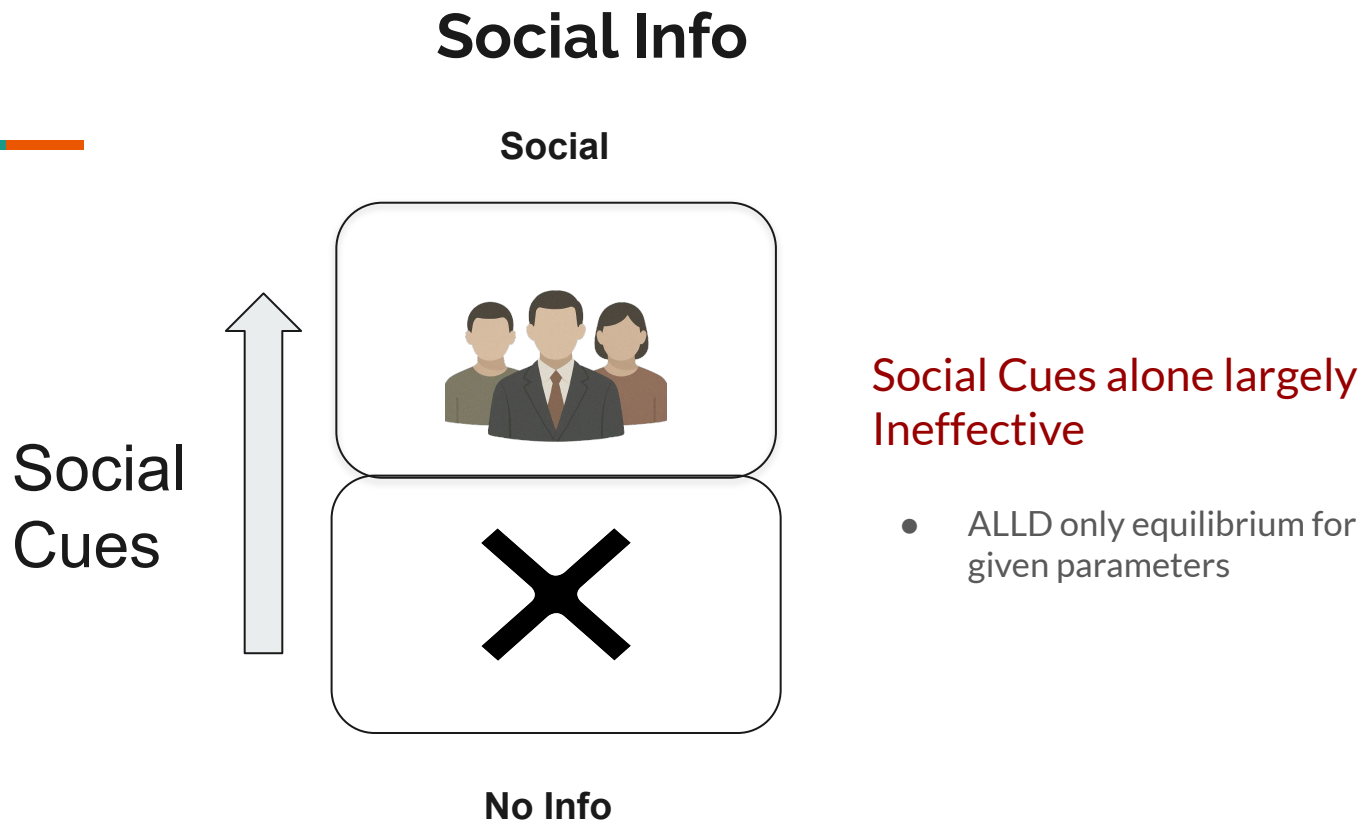
No information, no cooperation



No Info

- No cooperation

Summary



Local Stability Analysis



- All deterministic policies
- Epsilon perturbation
- Stable iff Final policy $\cong$ Initial policy

References



- Caring for the future can turn tragedy into comedy for long-term collective action under risk of collapse — Barfuss et al., PNAS (2020), 117(23), 12915–12922
- Modeling the effects of environmental and perceptual uncertainty using deterministic reinforcement learning dynamics with partial observability — Barfuss & Mann, Phys. Rev. E (2022), 105(3), 034409
- Deterministic limit of temporal difference reinforcement learning for stochastic games — Barfuss et al., Phys. Rev. E (2019), 99(4), 043305
- The collective-risk social dilemma and the prevention of simulated dangerous climate change — Milinski et al., PNAS (2008), 105(7), 2291–2294
- The effect of environmental information on evolution of cooperation in stochastic games — Kleshnina et al., Nat. Commun. (2023), 14, 39625
- Five rules for the evolution of cooperation — Nowak, Science (2006), 314(5805), 1560–1563
- pyCRLD: Collective Reinforcement Learning Dynamics in Python — Barfuss Lab, Simulation Software (2024), [GitHub](#)
-

Multi-agent Markov Decision Process

- **System State (s_t)** Description of the environment
- **Joint action space:**

$$\mathcal{A} = A_1 \times A_2 = \{(C, C), (C, D), (D, C), (D, D)\}$$

- **Transition Function (T):**

- $p \rightarrow g: P = \frac{N_p}{N} q_c$
- $d \rightarrow p: P = \frac{N_c}{N} q_r$

- **Reward Function:**

- If $s \& s' = p$:

$$R(C) = \frac{N_c f_c}{N} - c, \quad R(D) = \frac{N_c f_c}{N}$$

- else: $m < 0$

- **Observation Function (ω_r):**

$$o_t = \omega_r(s_t)$$

$$X_{(o)}^i = \begin{array}{c|cc} & \text{Cooperate} & \text{Defect} \\ \hline o_t^1 & P(C | o_t^1) & P(D | o_t^1) \\ o_t^2 & P(C | o_t^2) & P(D | o_t^2) \\ \vdots & \vdots & \vdots \\ o_t^n & P(C | o_t^n) & P(D | o_t^n) \end{array}$$

Return and Values

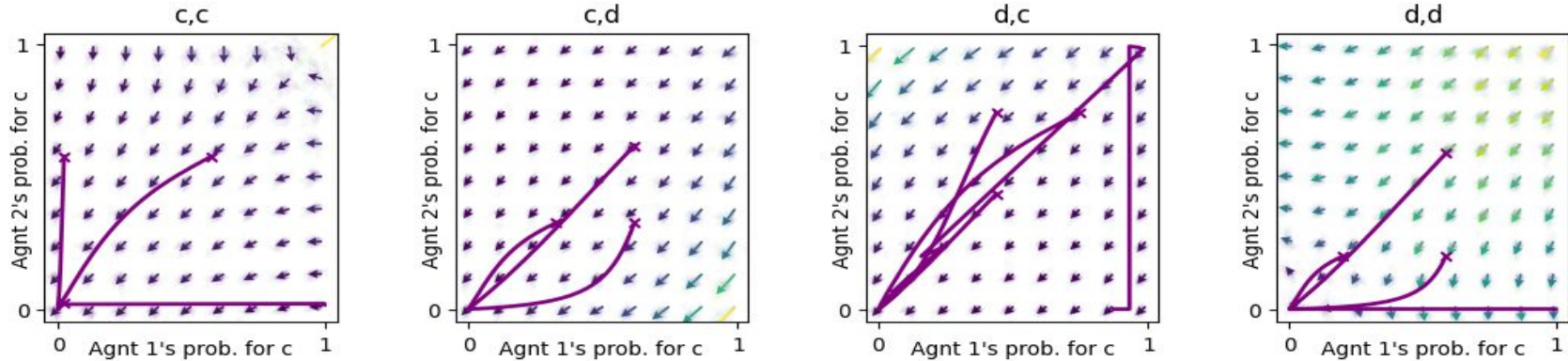

$$G^i(t) = (1 - \delta^i) \sum_{k=0}^{\infty} (\delta^i)^k r^i(t+k)$$

Return (Exponentially discounted sum of future rewards)

$$V_s^i(\mathbf{X}) = \text{TX} \langle G^i(t) \mid s(t) = s \rangle_s^i$$

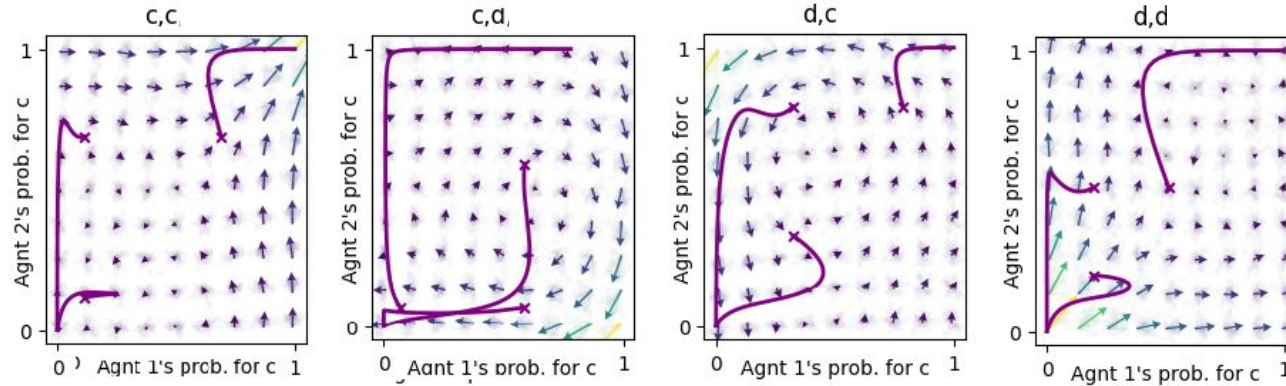
Value function (expected return starting from state “s”)

Only Social Info



- Mutual Defection with Minor Deviations

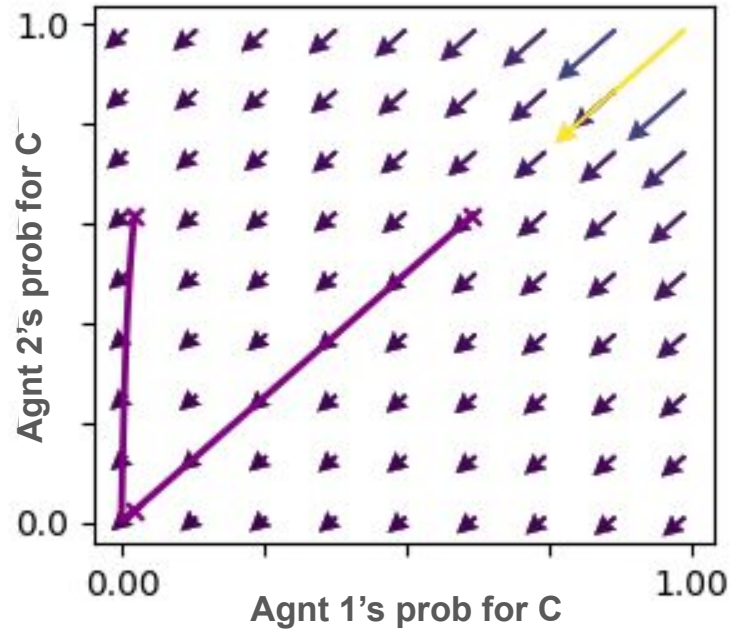
Both Social and Ecological Info



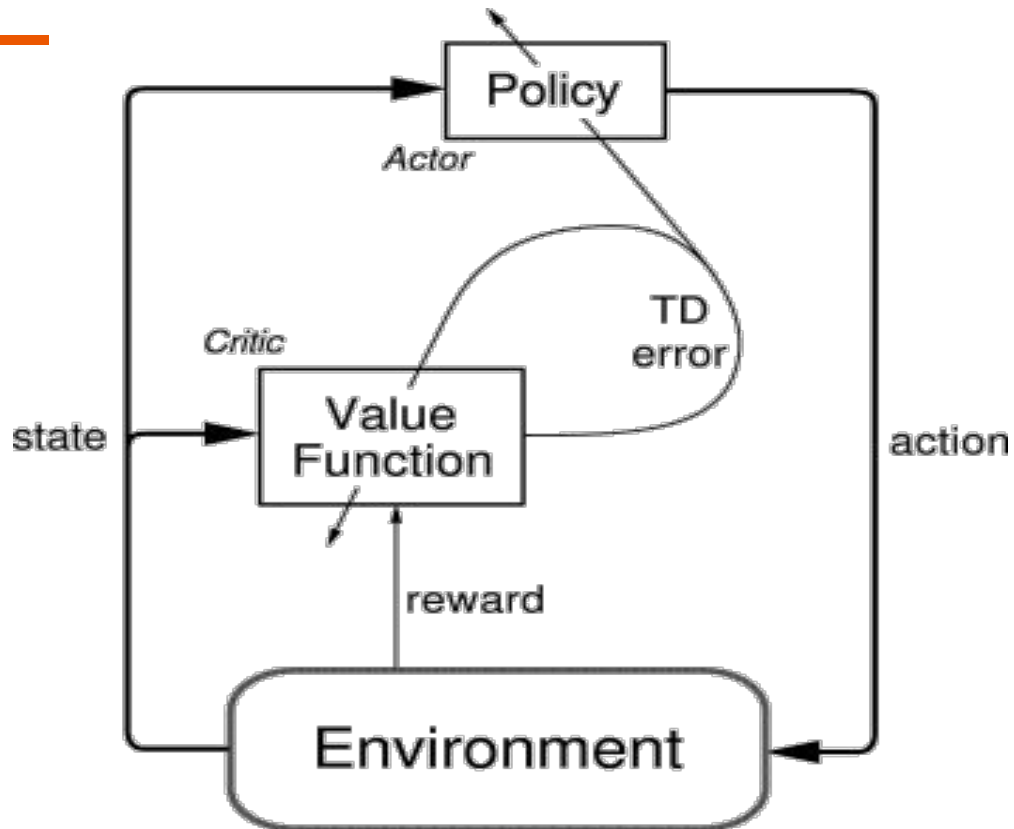
(No choice in degraded state)

- Cooperation Possible
- Complex Dynamics

No information

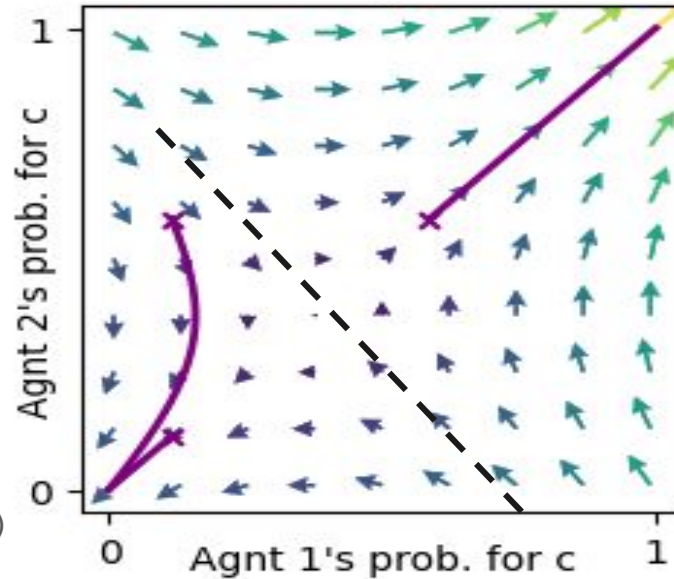


- Mutual Defection



- Reward-Based Policy Update
- Maximise cumulative returns

Only Ecological Info



Prosperous
State

(No choice in degraded state)

- Bistability

Learning Dynamics

- Temporal Difference Update:
$$D_{sa}^i(t) = \underbrace{(1 - \gamma^i)_{\text{TX-}i} \langle R \rangle_{sa}^i + \gamma_{\text{TX-}i}^i \langle V_{s'} \rangle}_{\text{new estimate}} - \underbrace{\tilde{V}_s}_{\text{earlier estimate}}$$
 - Critic: $\tilde{V}_s \leftarrow \tilde{V}_s + \alpha D_{sa}^i$
 - Actor: $X_{sa}^i(t+1) = \frac{X_{sa}^i(t) \exp[\alpha^i \beta^i D_{sa}^i(t)]}{\sum_b X_{sb}^i(t) \exp[\alpha^i \beta^i D_{sb}^i(t)]}$
- $\alpha \rightarrow$ Learning Rate
 $\beta \rightarrow$ Intensity of Choice
- Deterministic Limit:
 - $\lim_{\rightarrow \infty}$ Interactions before update (separation of timescales)



Potential Applications

- Clarifies how different informational cues shape cooperation in risk dilemmas.
- Helps guide monitoring policies by revealing which cues best support cooperation.

Existing Work



- Game-theoretic models and experimental studies (Tavoni and Iris 2020)

Assume perfect model of the environment

Evaluating Cooperation Levels

- Uniformly sample initial policies (N = 500)

- Compute final average cooperativeness:

$$\bar{p}_c = \frac{1}{2} \sum_i \sum_o p_{i,c}(o) \sigma_X(o)$$

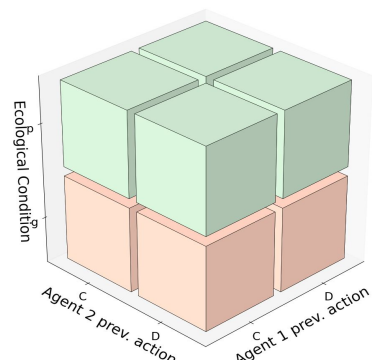
Probability of Cooperation

Stationary probability

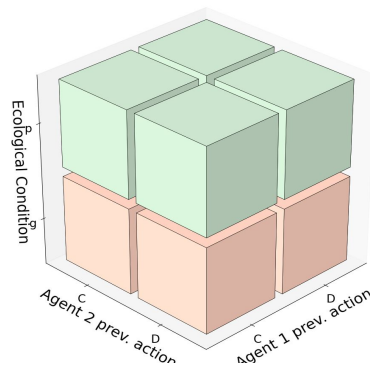
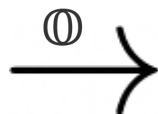
- “Cooperative” if $\bar{p}_c > 0.95$

* Only Prosperous States

Both Ecological & Social Information



System States

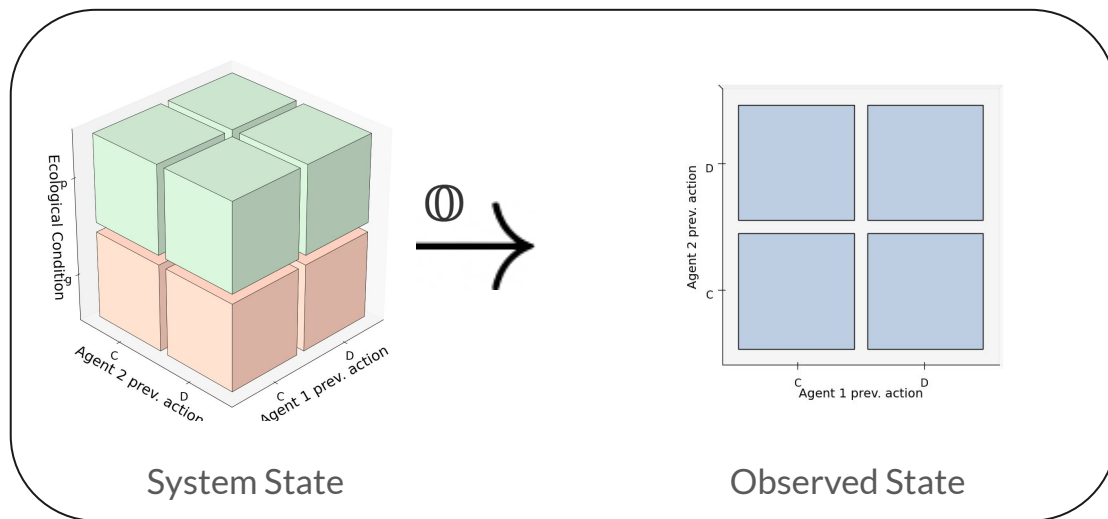


Observed States

Behaviour Profile

	Cooperate	Defect
p, C, D	$P(C \cdot)$	$P(D \cdot)$
p, D, C	$P(C \cdot)$	$P(D \cdot)$
p, C, C	$P(C \cdot)$	$P(D \cdot)$
p, D, D	$P(C \cdot)$	$P(D \cdot)$
g, C, D	$P(C \cdot)$	$P(D \cdot)$
g, D, C	$P(C \cdot)$	$P(D \cdot)$
g, C, C	$P(C \cdot)$	$P(D \cdot)$
g, D, D	$P(C \cdot)$	$P(D \cdot)$

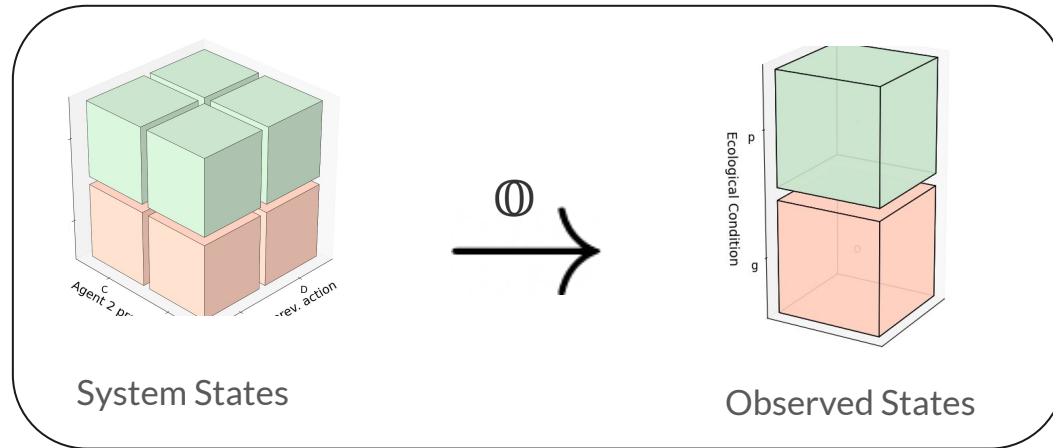
Only Social Information Regime



Policy

	Cooperate	Defect
C, D	$P(C \cdot)$	$P(D \cdot)$
D, C	$P(C \cdot)$	$P(D \cdot)$
C, C	$P(C \cdot)$	$P(D \cdot)$
D, D	$P(C \cdot)$	$P(D \cdot)$

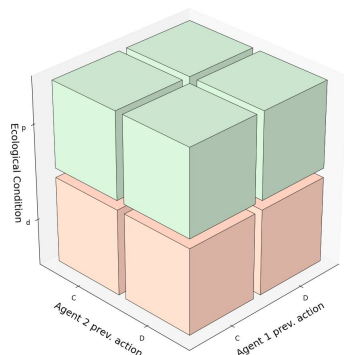
Only Ecological Information



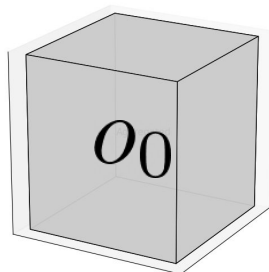
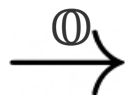
Policy

	Cooperate	Defect
p	$P(C .)$	$P(D .)$
g	$P(C .)$	$P(D .)$

Information Regime: No Information



System States



Observed States

Policy

	Cooperate	Defect
O_0	$P(C .)$	$P(D .)$